Exercise 2.2

1. For what values of natural number n, 4^n can end with the digit 6?
2. If m, n are natural numbers, for what values of m, does $2^n \times 5^m$ ends in 5?
3. Find the HCF of 252525 and 363636.
4. If $13824 = 2^a \times 3^b$ then find a and b.
5. If $p_1^{x_1} \times p_2^{x_2} \times p_3^{x_3} \times p_4^{x_4} = 113400$ where p_1, p_2, p_3, p_4 are primes in ascending order and x_1, x_2, x_3, x_4 are integers, find the value of p_1, p_2, p_3, p_4 and x_1, x_2, x_3, x_4.
6. Find the LCM and HCF of 408 and 170 by applying the fundamental theorem of arithmetic.
7. Find the greatest number consisting of 6 digits which is exactly divisible by 24,15,36?
8. What is the smallest number that when divided by three numbers such as 35, 56 and 91 leaves remainder 7 in each case?
9. Find the least number that is divisible by the first ten natural numbers.

2.5 Modular Arithmetic



Fig.2.5

In a clock, we use the numbers 1 to 12 to represent the time period of 24 hours. How is it possible to represent the 24 hours of a day in a 12 number format? We use 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 and after 12, we use 1 instead of 13 and 2 instead of 14 and so on. That is after 12 we again start from 1, 2, 3,... In this system the numbers wrap around 1 to 12. This type of wrapping around after hitting some value is called Modular Arithmetic.

In Mathematics, modular arithmetic is a system of arithmetic for integers where numbers wrap around a certain value. Unlike normal arithmetic, Modular Arithmetic process cyclically. The ideas of Modular arithmetic was developed by great German mathematician Carl Friedrich Gauss, who is hailed as the "Prince of mathematicians".

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Examples

1. The day and night change repeatedly.
2. The days of a week occur cyclically from Sunday to Saturday.
3. The life cycle of a plant.
4. The seasons of a year change cyclically. (Summer, Autumn, Winter, Spring)
5. The railway and aeroplane timings also work cyclically. The railway time starts at 00:00 and continue. After reaching 23:59, the next minute will become 00:00 instead of 24:00.

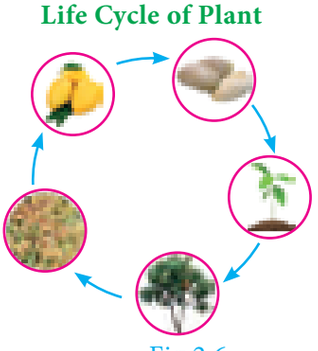


Fig.2.6

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